Helwan National University
Faculty of Humanities, Commerce, and Business Administration
Business Informatics & Digital Transformation (BIDT)
كلية العلوم الانسانية والتجارة وادارة الأعمال
برنامج معلوماتية الأعمال والتحول الرقمي

Proposal of

Final Year Graduation Project

{Project Name: Matgarco}

All-in-One Commerce Operating System

Project No. { 16 }

Team members

	Name	ID
1.	Salah Elden Haytham Abdelrahman.	971221180
2.	Joseph Adel Mosaad Basilius.	971221083
3.	Omar Zakaria Salem Soleiman.	971221223
4.	Amr Mohamed Ahmed Ebaid.	971221234
5.	Fawzy Khaled Fawzy.	971221239
6.	Youssef Diaa Eldeen Fekry Atia.	971221407

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1.1 Background

The global retail landscape is undergoing a significant paradigm shift, transitioning rapidly from traditional brick-and-mortar operations to a dynamic digital economy. Within the Egyptian market, this transition is strategically aligned with **Egypt Vision 2030**, which emphasizes "Digital Transformation" and "Financial Inclusion" as pivotal pillars for economic development. Empowering Small and Medium Enterprises (SMEs)—which represent the backbone of the local economy—to adopt digital technologies is no longer a luxury but a national necessity to ensure economic resilience and scalability.

Currently, the digital commerce sector in Egypt exhibits a distinct dichotomy. On one side, there is a heavy reliance on **"Informal Social Commerce,"** where a vast segment of merchants utilizes platforms like Facebook, Instagram, and WhatsApp as primary sales channels. While these platforms offer high reach, they lack the essential infrastructure for structured retail. Merchants in this category operate in a chaotic environment, suffering from manual order tracking via Direct Messages (DMs), lost customer inquiries, and the absence of a centralized customer database (CRM). This manual workflow creates severe operational bottlenecks, limits business scalability, and isolates merchants from the formal banking ecosystem.

On the other side, despite the availability of robust international e-commerce solutions (such as Shopify or Magento), a significant **"Technology & Usability Barrier"** prevents local adoption. The primary challenge is not merely financial cost, but the high **cognitive load** imposed on non-technical users. Many Egyptian merchants find complex dashboards, English-first interfaces, and technical configurations (hosting, domains, APIs) intimidating. Furthermore, the lack of seamless integration with local payment gateways (e.g., Fawry, Wallets) and local shipping providers forces merchants to resort to inefficient cash-based operations.

Consequently, there is a critical gap in the market for a solution that bridges the simplicity of social media with the power of professional e-commerce. A specialized platform is needed to "democratize" technology—offering a **Zero-Code**, localized environment that automates technical complexities while respecting the local operational culture. This project aims to fill this void, enabling merchants to transition from unorganized social selling to professional, data-driven digital enterprises.

1.2 Project Idea

The proposed project is a specialized SaaS (Software as a Service) platform designed to function as an **"All-in-One Commerce Operating System"** for SMEs in the MENA region. The core concept is to replace technical complexity with automated workflows, enabling non-technical merchants to launch and manage professional digital businesses without writing a single line of code.

Functionally, the system reimagines the store creation process. Instead of complex drag-and-drop editors, the platform utilizes a **"Smart Onboarding Wizard."** This interactive module guides the merchant through a structured data-collection process (business type, branding preferences, and catalog setup). Based on these inputs, the system's engine automatically generates a fully optimized, high-performance **Progressive Web App (PWA)** storefront. This ensures a native-app-like experience for end-users while maintaining a standardized, high-quality codebase for the merchant.

Beyond the storefront, the project provides a unified **Merchant Command Center.** This backend dashboard orchestrates the entire retail lifecycle by centralizing inventory tracking, order fulfillment, and customer data. It acts as an integrated ecosystem that connects seamlessly with local payment gateways and shipping providers via APIs, effectively solving the fragmentation issues inherent in social commerce.

Furthermore, the system distinguishes itself through an embedded **Artificial Intelligence Module** that serves two strategic roles:

1. **Content Assistant (Generative AI):** Reduces operational friction by automatically generating SEO-friendly product descriptions and marketing content, helping merchants populate their stores rapidly.
2. **Strategic Advisor (Analytical AI):** Provides decision support by analyzing sales patterns to recommend inventory restocks and suggest personalized product bundles, thereby transforming raw data into actionable business intelligence.

1.3 Project Scope

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The scope of this project defines the specific boundaries of the **Matgarco** platform, outlining the deliverables to be implemented and the limitations set for the current phase.

1. In-Scope (System Deliverables):

- **Merchant Dashboard (SaaS Core):** Developing a comprehensive control panel allowing merchants to subscribe, manage products, track inventory, and view sales analytics.
- **Customer Storefronts:** Generating dynamic, responsive web interfaces for end-users to browse products, add items to the cart, and proceed to checkout securely.
- **Super-Admin Dashboard:** A centralized management interface for platform administrators to manage merchant subscriptions, monitor platform revenue, and handle system-wide settings.
- **Third-Party Integrations:** Integrating with local payment gateways (e.g., Fawry , Paymob) and shipping APIs to automate delivery cost calculations and order tracking.
- **AI-Powered Features:** Implementing lightweight AI tools to assist merchants in generating Arabic product descriptions and optimizing store content.

2. Out-of-Scope (Exclusions):

- **Native Mobile Applications:** The project focuses on developing a high-performance **Web Application (PWA)** accessible via mobile browsers. Native Android (Kotlin) or iOS (Swift) apps are excluded from this phase.
- **Payment Gateway Development:** The system **integrates** with existing authorized payment providers; it does not involve building a financial processing engine or banking system from scratch.
- **Physical Logistics Operations:** The platform manages shipping data and tracking via APIs but does not involve managing physical delivery fleets, drivers, or warehousing logistics.
- **Content Production:** Merchants are solely responsible for uploading their specific business data (product photos, descriptions, and branding assets).

1.4 Problem Statement

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The core problem addresses the **"Digital Stagnation & Operational Fragility"** facing Small and Medium Enterprises (SMEs) in the local market. Despite the high demand for e-commerce, these merchants are currently trapped in a **"Retail Limbo"** between two unviable extremes:

1. The Inefficiency of Informal Social Commerce:

Merchants relying solely on social media channels (Facebook/Instagram) suffer from severe operational bottlenecks:

- **Lost Revenue:** High volumes of customer inquiries in comments and DMs lead to missed sales opportunities due to slow manual response times.
- **Data Blindness:** Merchants do not own their customer data; it remains locked within the social platforms, preventing retargeting or building loyal customer bases.
- **Trust Deficit:** The lack of a professional, branded storefront creates skepticism among modern consumers regarding credibility and payment security.

2. The Barrier of Global Solutions:

Existing enterprise platforms (e.g., Shopify, wuilt) fail to serve this segment due to a "Barriers to Entry" crisis:

- **Financial Instability:** Reliance on foreign currency (USD) for monthly subscriptions makes costs unpredictable and often unsustainable for local businesses.
- **Cognitive Overload (Technical Barrier):** The complexity of standard dashboards, English-first interfaces, and the requirement for technical configuration (DNS, Hosting) intimidates non-technical merchants, causing them to abandon digital transformation efforts.

3. The Cost of Inaction (Scalability Crisis):

Without a specialized solution, these merchants face an existential threat. They remain **"Unscalable,"** unable to handle growth manually. Consequently, they risk losing significant market share to digitally mature competitors **("Market Whales")** who utilize automated systems, eventually leading to their marginalization.

1.5 System Objectives

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Main Objective:

To architect and deploy a specialized, zero-code SaaS ecosystem that democratizes digital commerce for SMEs in Egypt. The primary goal is to lower the technical entry barrier, enabling non-technical merchants to launch professional, scalable online stores in under 10 minutes, thereby facilitating a seamless transition from informal social selling to structured digital enterprise.

Specific Objectives:

1. Eliminate Technical Friction (Zero-Code Empowerment):

To design an intelligent "Wizard-Based" onboarding engine that abstracts complex backend processes (hosting, deployment, database setup). This allows users to generate fully functional Progressive Web Apps (PWAs) solely through guided business logic inputs, achieving a **"Zero-Code"** experience.

2. Orchestrate and Automate Operations:

To develop a unified "Central Command Dashboard" that resolves operational fragmentation by:

- Centralization: Aggregating inventory, orders, and customer data into a single source of truth.
- Logistics Automation: Integrating directly with third-party shipping APIs to automate courier booking and delivery tracking, replacing manual coordination.

3. Implement AI-Driven Intelligence:

To integrate Artificial Intelligence modules that enhance both productivity and strategy:

- Generative AI (Content Automation): To utilize Natural Language Processing (NLP) for instantly generating SEO-ready product descriptions, solving the "content bottleneck."
- Analytical AI (Decision Support): To apply data analytics algorithms that provide actionable insights on sales trends and customer behavior to drive revenue growth.

4. Ensure Market Localization & Financial Inclusion:

To tailor the system architecture specifically for the MENA region by:

- Cultural Localization: Providing a native Arabic-first interface and user experience (UX).
- Fintech Integration: Bridging the gap between cash and digital payments by integrating trusted local payment gateways and e-wallets, promoting financial inclusion for unbanked merchants.

1.6 System Features

The **Matgarco** platform provides a comprehensive suite of features structured around three primary user roles, ensuring a robust and scalable e-commerce ecosystem.

A. Merchant Module (The Business Hub)

- **Store Customization:** Merchants can select from professional, responsive themes and customize branding assets (Logo, Colors, Fonts) to build a unique identity.
- **Team & Role Management:** A secure access control system allowing merchants to create **sub-accounts for employees** with specific permissions (e.g., "Order Manager" vs. "Inventory Editor") to protect sensitive data.
- **Product & Inventory System:** Full control to add unlimited products, manage variants (Size/Color), categorize items, and monitor stock levels in real-time.
- **Marketing Engine:** Tools to generate **Discount Coupons** and Promo Codes to boost sales and customer retention.
- **Advanced Analytics:** Interactive dashboards displaying critical KPIs, including total revenue, best-selling products, and visitor demographics.
- **Order Fulfillment:** A streamlined workflow to track order statuses (Pending, Shipped, Delivered), print shipping labels, and manage returns.
- **Support Center:** A built-in ticketing system for merchants to report technical issues directly to platform administration.

B. Customer Module (The Storefront)

- **Bilingual Experience:** A fully localized interface supporting both **Arabic and English** to cater to a wider audience.
- **Engagement & Social Proof:** Features allowing customers to rate products, write reviews, and save items to a **Wishlist**.
- **Smart Search & Filtering:** High-performance search functionality with filters for price range, category, and specifications.
- **Secure Checkout:** A simplified, secure purchase process supporting local payment gateways (Cards/Wallets) and Cash on Delivery (COD).
- **Live Order Tracking:** Customers can track their shipment status via a dedicated tracking page.

C. Super-Admin Module (Platform Control)

- **Subscription Management:** Comprehensive tools to manage merchant plans (Free, Basic, Pro), billing cycles, and renewals.
- **System Health Monitoring:** A master dashboard to oversee platform performance, total active stores, and aggregate revenue.
- **User Governance:** Ability to suspend or ban accounts violating platform terms and manage support tickets.

D. Core Services & AI Integration

- **Automated Notifications:** A system that sends automatic email/SMS updates to customers (Order Confirmation, Shipping Update) and merchants (New Order Alert).
- **AI Content Assistant:** Automated generation of professional Arabic product descriptions and SEO-friendly titles to assist merchants.
- **Localized Logistics:** Integration with shipping APIs to calculate delivery fees instantly based on the customer's governorate.

1.7 Related Work and Comparative Study

To validate the necessity of the proposed system, a comprehensive analysis of the current e-commerce landscape was conducted. The study aims to identify the limitations of existing platforms and define the "Research Gap" that this project addresses. The market solutions were categorized into **Global SaaS Leaders** and **Regional Competitors**.

1.7.1 Analysis of Existing Solutions

A. Global Market Leaders (e.g., Shopify, Wix, WooCommerce):

These platforms represent the global standard for e-commerce.

- **Critique:** While technically robust, they suffer from a "Generalization Problem." They are designed for a universal market, lacking native integration with Egypt's local fintech (e.g., Fawry, InstaPay) and logistics networks. Furthermore, the reliance on USD-based pricing creates financial unpredictability for local SMEs. WooCommerce, specifically, imposes a high "Technical Maintenance Burden" (hosting, SSL, updates) that alienates non-technical users.

B. Regional Competitors (e.g., Salla, Zid):

These platforms have successfully localized the experience for the Gulf (GCC) market.

- **Critique:** While they solve the language barrier, their pricing models are often tailored to higher-income economies. More importantly, their current architecture focuses on "Store Management" rather than "Content Automation." They lack deep integration of Generative AI for solving the "Empty Store Syndrome" (writing descriptions and marketing copy), which remains a manual hurdle for merchants.

1.7.2 The Identified Research Gap

The analysis reveals a critical void in the market. There is currently no solution that simultaneously offers:

1. **Hyper-Localization:** (EGP pricing + Egyptian Logistics).
2. **Zero-Code Simplicity:** (No technical setup required).
3. **Native AI Automation:** (Generative content creation as a core feature, not a plugin).

1.7.3 Comparative Matrix

The following table contrasts the proposed system against market standards to highlight the competitive advantage.

Table 1: Comparative Analysis Matrix

Feature / Criteria	**Shopify (Global)**	**WooCommerce (Open Source)**	**Salla/Zid (Regional)**	**Proposed System**
Pricing Model	USD Subscription (Volatile)	Free (High Hosting/Dev Costs)	High-Tier Subscription	**EGP Subscription (Stable)**
Technical Entry Barrier	Moderate (Complex UI)	**High** (Requires Dev Skills)	Low	**Zero (Wizard-Based)**
Logistics Integration	Manual / Paid Plugins	Manual Configuration	Native (GCC Focus)	**Native (Egypt Focus)**
AI Content Generation	3rd Party Apps (Extra Cost)	Plugins Only	Limited	**Core Native Feature**
Target User	Tech-Savvy Merchant	Developer / Agency	Established Business	**Non-Tech SME**

1.8 Project Planned Price

1.8 Project Planned Price and Financial Structure

This section outlines the financial and pricing framework of the proposed system, including subscription tiers, transaction-based commissions, and long-term revenue sustainability. The pricing strategy is designed to balance accessibility for small merchants with predictable recurring revenue, while remaining competitive within the Egyptian market.

1.8.1 Revenue Model (Hybrid Pricing Strategy)

The platform adopts a Hybrid Pricing Strategy that combines:

- **Transaction-based commissions for early-stage merchants**
- **Fixed monthly and annual subscriptions for growing and established businesses**

This strategy allows merchants to join the platform according to their business maturity, while ensuring continuous platform monetization.

Pricing Plans Overview

The system provides four access levels:

1. **Free Trial**
2. **Starter Plan (Commission-Based)**
3. **Professional Plan (Fixed Subscription)**
4. **Business Plan (Advanced Subscription)**

Plan 0: Free Trial

Purpose:
Allow new merchants to test the platform's core features before financial commitment.

Duration:

- **14 days free trial**

Pricing Structure:

- **Subscription Fee: Free**
- **Transaction Fee: 3% per successful transaction**

Included Features:

- **Basic online store setup**
- **Limited product listings**
- **Order and customer management**
- **Basic sales dashboard**
- **Platform-branded subdomain**
- **Standard customer support**

Limitations:

- **No advanced analytics**
- **Limited AI usage**

This plan reduces entry barriers and increases merchant onboarding rates.

Plan 1: Starter Plan

Target Segment:
Micro and small merchants entering structured e-commerce.

Pricing:

- **Monthly Subscription: 250 EGP / Month**
- **Transaction Fee: 2% per successful transaction**

Annual Plan Discount:

- **2,500 EGP / Year (Save ~17%)**

Included Features:

- **Full store customization**
- **Increased product limits**
- **Integrated local payment gateways**
- **Shipping integration**
- **AI-generated product descriptions (limited credits)**
- **Standard analytics and sales reports**

Value Proposition:
Provides affordability while monetizing low-volume merchants fairly through commission.

Plan 2: Professional Plan

Target Segment:
Growing merchants with consistent sales volume.

Pricing:

- **Monthly Subscription: 450 EGP / Month**
- **Transaction Fee: 0%**

Annual Plan Discount:

- **4,500 EGP / Year (Save ~17%)**

Included Features:

- **Unlimited product listings**
- **Advanced sales analytics and performance insights**
- **Higher AI content generation limits**
- **Custom domain mapping**
- **Discount and coupon management**
- **Priority customer support**

Value Proposition:
Encourages business growth without transaction-based revenue sharing.

Plan 3: Business Plan

Target Segment:
High-volume sellers, brands, and professional operations.

Pricing:

- **Monthly Subscription: 699 EGP / Month**
- **Transaction Fee: 0%**

Annual Plan Discount:

- **6,999 EGP / Year (Save ~17%)**

Included Features:

- **Multi-user staff access and role management**
- **Advanced inventory forecasting and alerts**
- **AI-powered sales and marketing recommendations**
- **Exportable financial reports (PDF / CSV)**
- **API access for integrations (future-ready)**
- **Dedicated support channel**

Value Proposition:
Designed for merchants seeking full operational control and data-driven scalability.

1.8.2 Estimation of Start-up Capital

Following the standard business planning framework, the project's valuation is categorized into Fixed Investments and Working Capital4.

A. Investment (Fixed Assets & Infrastructure):

These are the one-time costs required to establish the technical foundation.

- **Equipment (Hardware):** High-performance workstations for development (Virtual Asset provided by team).
- **Digital Assets:** Domain Name Registration (.com) and SSL Certificates (~$15/year).
- **Software Licenses:** Integrated Development Environments (IDEs) and Design Tools (Figma).
- **Total Investment:** Estimated at **20,000 EGP.**

B. Working Capital (Operational Costs):

These represent the operational expenses required to sustain the project during the 6-month development phase5.

- **Staff Costs (Virtual Valuation):** The theoretical salary for the 6-member cross-functional team (Development, Analysis, Design) over 6 months.
    - *Calculation:* 6 members × 8,000 EGP × 6 Months = **288,000 EGP.**
- **Operational Costs:** Server hosting (VPS), AI Model inference computational power, and internet connectivity (~$20/month).
- **Marketing & Promotion:** Initial budget for social media awareness campaigns.
- **Total Working Capital:** Estimated at **300,000 EGP.**

1.8.3 Sources of Start-up Capital

Since this is a graduation project transitioning into a startup, the funding structure relies on internal equity6.

- **Equity Capital:** 100% "Bootstrap" funding. The initial actual costs (hosting/domain) are covered by the **Partners' Own Savings** (the team members).
- **Sweat Equity:** The technical development effort (valued at 288,000 EGP) is invested by the team as intellectual property capital.

1.9 Project Planned Place

1.9 Project Planned Place (Deployment & Market Scope)

In alignment with the digital nature of the project, the "Place" strategy is defined by three dimensions: the Hosting Infrastructure (where the data lives), the Target Market (where the users live), and the Distribution Channel (how the service is delivered).

1. Hosting & Infrastructure Location:

The system backend and database will be hosted in European Data Centers (e.g., Germany) using cloud providers like Contabo or Hetzner.

- Strategic Reasoning: Hosting in Europe ensures the lowest network latency (fastest response time) for Egyptian users compared to US-based servers, while maintaining strict data security standards.

2. Target Geographical Market (Phased Rollout):

The project follows a structured expansion roadmap to manage scalability risks:

- **Phase 1 (The Anchor Market - Egypt):** The initial launch targets Egyptian SMEs to validate the "Market Fit," utilizing local payment gateways (Fawry) and shipping integrations (Bosta).
- **Phase 2 (Regional Expansion - MENA):** Scaling to key Arab markets, specifically Saudi Arabia (KSA) and UAE, capitalizing on the Arabic-first architecture.
- **Phase 3 (Global Vision):** Long-term plans to enter international markets as a competitive cross-border commerce tool.

3. Distribution Channels:

The service utilizes a Direct-to-Consumer (DTC) Web Distribution model.

- Accessibility: The platform is accessible 24/7 via any web browser as a Progressive Web App (PWA). This eliminates the need for physical branches or third-party distributors, allowing merchants to "subscribe and start" instantly from their homes.

4. Development Environment (Operational Base):

The project development executes within a Remote/Hybrid Agile Workspace. The team operates as a distributed workforce using collaborative tools (Jira, GitHub, Discord) to simulate the operational dynamics of modern global tech startups.

1.10 Project Planned Promotion

1.10 Project Planned Promotion (Go-to-Market Strategy)

The promotion strategy is rigorously designed based on the **AIDA Model** (Attract Attention, Create Interest, Arouse Desire, Call for Action) as recommended in standard marketing frameworks. The strategy shifts from traditional advertising to an "Inbound Marketing" approach, focusing on education and trust-building.

1. Awareness & Education (Attracting Attention):

To address the "Attract Attention" phase2, the project will utilize Content Marketing to bridge the technical knowledge gap:

- **Short-Form Educational Video:** Leveraging platforms like TikTok and Reels to publish "Micro-Learning" content (e.g., "How to start selling online in 5 minutes"). This aligns with providing easy access to product information.
- **Community Penetration:** Active engagement in niche Facebook groups for wholesalers. This strategy falls under **"Publicity and Public Relations"**, positioning the brand as a helpful industry leader rather than just a vendor.

2. Viral Growth Loops (Creating Interest):

To "Create Interest" and maximize reach with minimal cost:

- **The "Powered-By" Signature:** A growth hack where every store launched on the free tier displays a clickable "Powered by [Project Name]" footer.
- **Referral System:** Implementing a reward-based referral program (e.g., "Invite a merchant, get 1 month free"). This effectively utilizes **"Personal Selling"** dynamics by turning satisfied users into promoters.

3. Strategic Partnerships (Arousing Desire):

To "Arouse Desire" and build credibility:

- **Logistics Partners:** Collaborating with trusted shipping providers (e.g., Bosta) to offer the platform as a recommended solution.
- **Influencer Marketing:** Providing "Pro Accounts" to business influencers. This serves as a powerful testimonial mechanism to validate the product's value.

4. The Entry Hook (Call for Action):

To trigger the final "Call for Action" and overcome purchase hesitation:

- **Risk-Reversal Offer:** Offering a **14-Day Free Trial** with no credit card required. This directly implements the concept of using **"trial offers"** to remove barriers to entry.

1.11 System Requirements

Following the project analysis, the system requirements are categorized into functional requirements (core behaviors) and non-functional requirements (quality attributes).

A. Functional Requirements (What the system must do)

- The system must allow merchants to **register** ,create business profiles, and manage subscription plans securely.
- The system must enable merchants to perform **data entry** for product catalogs and manage inventory updates in **real-time.**
- The system must generate automated **sales reports** and revenue analytics accessible via the admin dashboard for decision-making.
- The system must facilitate secure **payment processing** by integrating with local gateways and handling the complete checkout workflow.
- The system must allow administrators to approve user accounts, monitor system logs, and resolve technical support tickets.

B. Non-Functional Requirements (Qualities of the system)

- **Performance :**The system must ensure fast response times, with web pages loading in **under 3 seconds** to maintain user engagement.
- **Availability :**The system must be available **7/24** with minimum downtime to ensure continuous e-commerce operations.
- **Security :**The system must **store data securely** ,encrypting sensitive user information and ensuring safe data transmission via **HTTPS.**
- **Usability :**The interface must be **simple** and intuitive for non-technical users, fully supporting the Arabic language.
- **Scalability :**The system architecture must be scalable to handle a growing number of concurrent users and transactions without performance degradation.

1.12 System Users

User Type	Description	Purpose
Customers (End-Users)	Individuals visiting the online storefronts to purchase products.	Search for items, place orders, track shipments, and provide ratings/reviews.
Merchants (Business Owners)	Small business owners who subscribe to the SaaS platform to manage their online presence.	Create the store, manage subscriptions, monitor revenue, and oversee business performance.
Store Staff	Employees hired by the merchant (e.g., Inventory Managers or Sales Reps).	Assist in daily operations like updating stock levels and processing orders with limited permissions.
Super Administrators	The technical and management team responsible for the Matgarco platform.	Manage the entire system, monitor platform health, handle merchant subscriptions, and resolve support tickets.

1.13 System tools

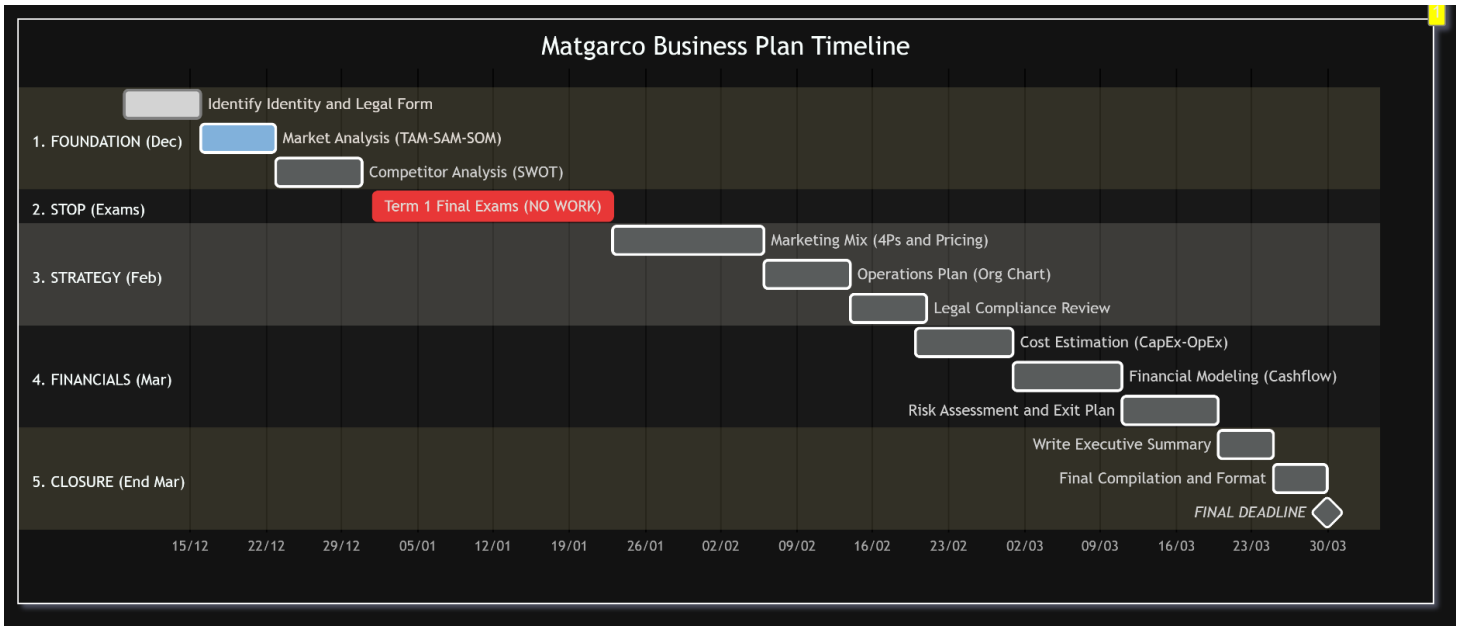
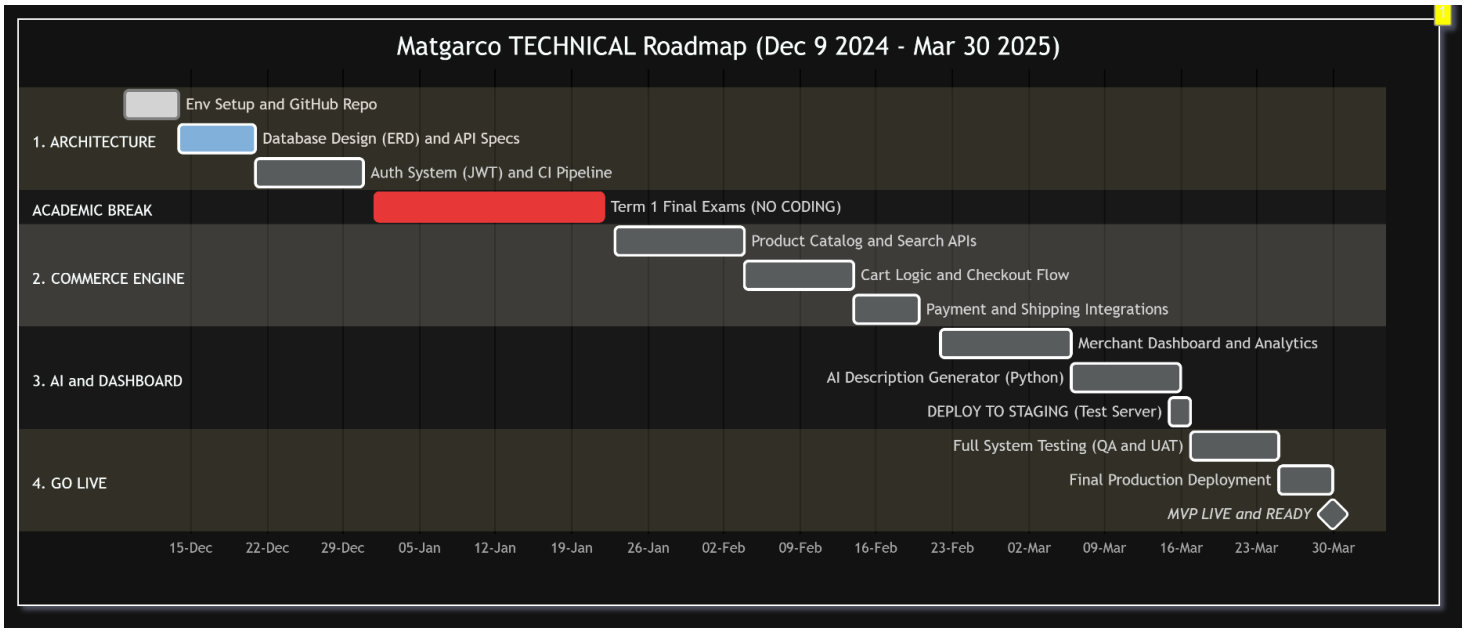
1.13 System Tools

The following tools and technologies will be used for research, design, development, and project management:

- **Documentation :**Notion (for meeting notes and technical documentation).
- **Project Management :**Jira (for **Agile** task tracking and sprint planning).
- **Business Modeling :**Draw.io (for creating ERDs, Use Case diagrams, and Flowcharts).
- **Research Tools :**Google Scholar (for academic research and comparative studies).
- **Interface Design :**Figma (for UI/UX prototyping and wireframing).
- **Frontend Development :**HTML5, CSS3, JavaScript, and React.js framework.
- **Backend Development :**Node.js environment with Express.js framework.
- **Database Management :**MongoDB (NoSQL).
- **AI Engine:** Python (FastAPI) & Hugging Face Transformers (Running localized Open-Source LLMs for zero-cost generation).
- **API Testing :**Postman (for testing and debugging RESTful APIs).
- **Version Control :**Git & GitHub (for source code management).
- **Development Tools :**Visual Studio Code (VS Code).
- **Hosting Platform :**Localhost for initial development, transitioning to Cloud Hosting for final deployment.

1.14 Project time plan

- **Matgarco Business Time Plan:**



- **Matgarco Technical Time Plan:**